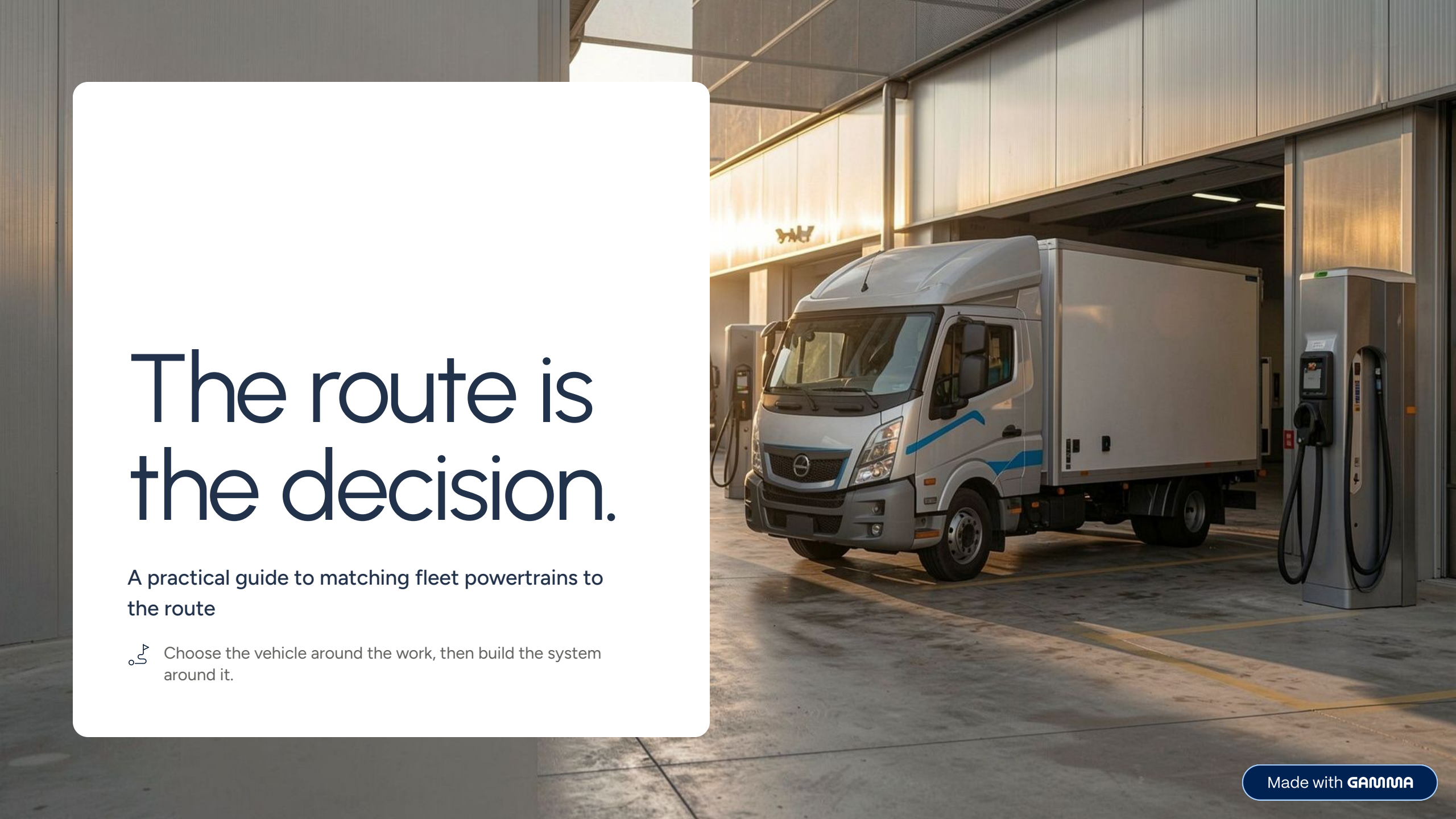
# The route is the decision.

## A practical guide to matching fleet powertrains to the route

Choose the vehicle around the work, then build the system around it.



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# Pick the powertrain that fits the work

The route decides what “fit” means.



## Depot-based and predictable

Return-to-depot routes favor battery-electric trucks, where charging can be planned around the operating day.



## Long-haul and high-utilization

Hydrogen or liquid-fuel pathways may preserve distance, payload, and utilization where batteries are harder to fit.



## Dense urban drops

Smaller, lighter, more agile vehicles solve a different operating problem at the curb.

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# Electric does the everyday heavy lifting

BEST FIT

## Return-to-depot routes

Regional delivery and predictable daily mileage make charging and dispatch easier to coordinate.

STRENGTHS

## Efficient, quiet, clean at the tailpipe

High drivetrain efficiency, quiet operation, and zero tailpipe emissions.

WATCH-OUTS

## Infrastructure is part of the vehicle

Plan for charging capacity, dwell time, payload trade-offs, and winter performance.





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# Hybrid is the bridge — use it on purpose

Use the bridge deliberately, then know when to leave it.

01

### 1 · Mixed routes

Useful when charging access is uncertain or a fleet is beginning a staged transition.

02

### 2 · Reduce engine use

As charging access and route confidence improve, shift more of the work to electric operation.

03

### 3 · Set the trigger

Treat the hybrid as a bridge with a clear replacement trigger, not the final state.

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# Hydrogen steps in when payload wins

- Best fit: long distances, high payloads, and fast refueling requirements
- Fuel-cell trucks can preserve operating patterns that are harder to electrify with batteries
- Economics depend on reliable access to low-carbon hydrogen and fueling infrastructure





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# Biofuels keep today moving — with trade-offs

A familiar bridge can still create a long-term decision.

BEST FIT

## Existing diesel assets

Lower-carbon bridge options can reduce disruption where replacement is not yet practical.

ADVANTAGES

## Keep the system familiar

Vehicles, fueling systems, and operating practices remain recognizable to the team.

EVALUATE BEFORE SCALING

## Familiar does not mean risk-free

Check feedstock availability, lifecycle emissions, supply certainty, and long-term lock-in.

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# Cargo bikes own the last 500 meters

## Best fit

Dense urban zones, short drops, congestion, and curb-constrained streets.

## Operating pattern

Shift parcels to a microhub, then complete deliveries with cargo bikes.

## Design the handoff

Plan for payload, weather, battery swaps, rider safety, and secure handoff points.

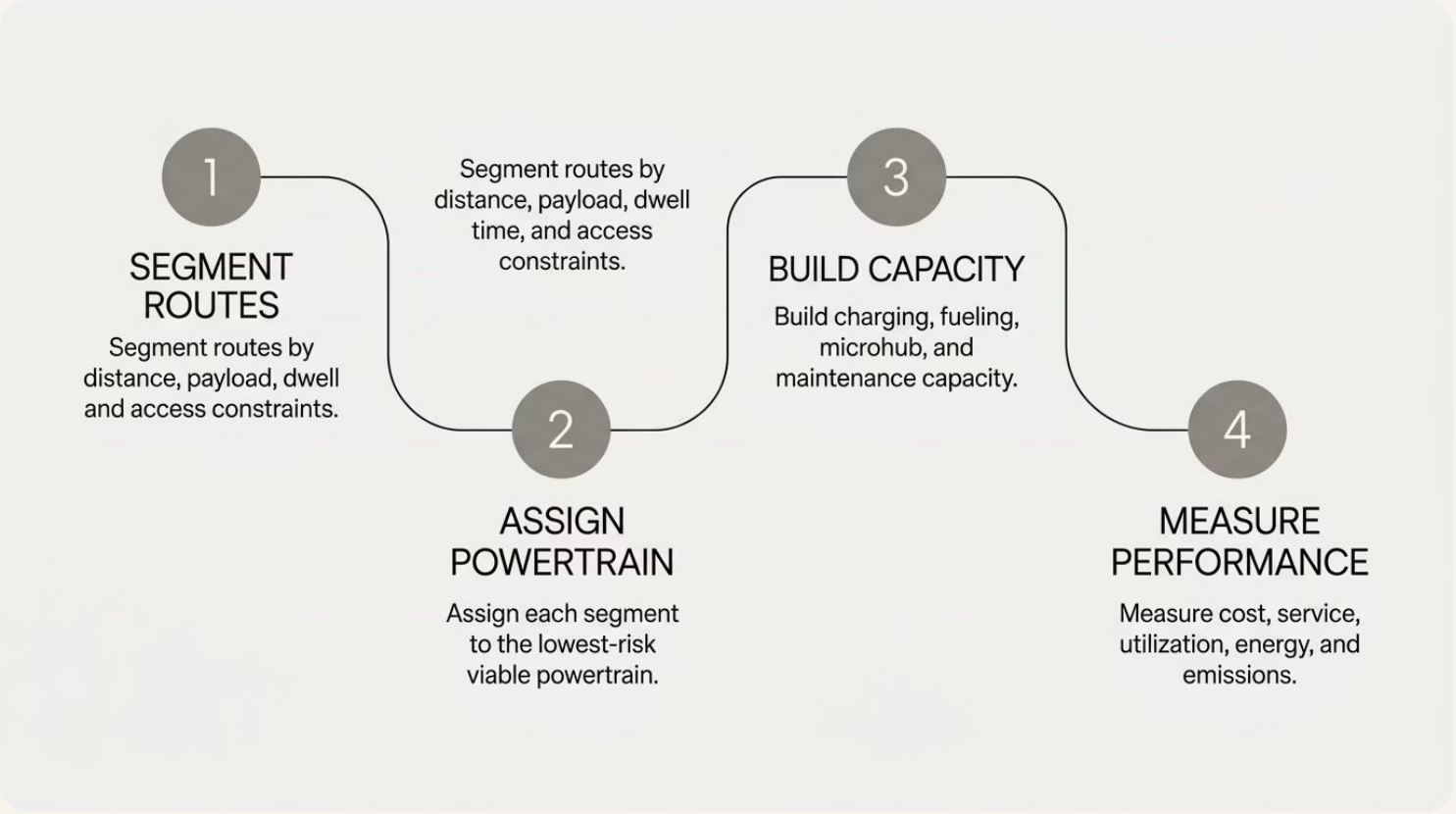
Evidence base: World Economic Forum reporting on urban delivery models.



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# New vehicles need a new operating model

Powertrain selection is the start of a system decision.




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# Pilot the shift. Prove the system.

Replace assumptions with a pilot that reflects the work.

### Baseline

Routes, mileage, payload, stops, dwell, fuel, and service levels.

### Pilot

A representative depot, a demanding regional route, and one urban zone.

### Scale

Expand only when total cost, uptime, and customer experience clear the bar.

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# Match the route. Build the system.

Replace diesel where the route is ready, preserve optionality where it is not, and make infrastructure the scaling plan.



The practical transition is a portfolio of fit-for-route choices.

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